Machine Learning for Professional Chess Match Outcome Prediction

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Abstract. Predicting the outcomes of chess games or any other sporting event is a difficult task due to the number of factors that come into play both before and during a game. This Project therefore aims to determine whether a game of chess can be predicted using data collected from online chess platforms. Within the proposal a brief introduction is given into the background of chess and machine learning and why machine learning could play a vital role in identifying obscure factors that contribute to a win. Additionally, a detailed project plan has been put forward to map the requirements and scope of the project along with a risk assessment to manage and tackle risks moving forward.

Keywords: Machine Learning, Chess Prediction, Predictive Modeling, Data Collection, Data Preprocessing

1 Introduction

1.1 Chess

The history of the game, chess, can be traced back over 1,500 years to its routes in Inda in the 6th century CE. Since then, chess has seen many changes until the modern rules of chess were instantiated during the 15th century in Italy and Spain. However, it wasn't until the 19th century that competitive chess emerged, with early tournaments in Europe taking place with the first chess championship in 1886. Since this time, players from all over the globe have studied strategies in the hopes of finding new innovative tactics to advance the game and see them reach the pinnacle of competitive chess (Dvorsky, 2015; House of Staunton, n.d.).

In 1997, Garry Kasparov, a world chess champion faced a new type of competitor in the form of International Business Machines (IBM) Deep Blue. This event marked the first time a machine had beaten a world chess champion and in turn made a significant impact in the chess world, changing how players both study and prepare for games (Dvorsky, 2015). At present, chess players and their teams have the capability to analyze games using chess engines enabling them to draw insights from vast historical databases to seek any possible edge over their opponents.

1.2 Machine Learning

Machine learning is a sub field of Artificial Intelligence that in its most basic form has the aim to enable computers to learn by creating models mapped from a given set of data inputs in the hopes of returning a "correct" predicted output (Bi et al., 2019). Machine learning has become a common tool used by scientists and engineers alike to gain insights from the wealth of data being collected in today's computing world. Examples of machine learning can be found in many industries catering for the expanse of data at their fingertips, from predicting best treatments for patients based on medical records to improving traffic control operations to reduce congestion (Jordan and Mitchel, 2015).

1.3 Machine Learning for Chess Match Prediction

This Project aims to create a multiclass classification machine learning model using supervised learning that can accurately predict a win, loss or draw between two professional chess players. Therefore, to accomplish this aim a historical chess dataset with labels for win, loss and draws will need to be created from the vast amounts of games played online. The idea of this project centered around game match prediction however is not new and will join the wealth of prior research carried out around competitive sports match prediction using machine learning methods (Langaroudi and Yamaghani, 2019, 30-33; Bunker and Susnjak, 2022, 1290-1305). Alongside this, chess match prediction has been tried previously but with more emphasis on game strategy such as positioning and theoretical openings using the chess board to obtain insights (Dreżewski and Wątor, 2021; Brown et al., 2017). For this project the emphasis will be on the external features rather than solely match features, for example, less focus on individual moves made in a current position but rather factors such as average move time or current player form (Bunker and Thabtah, 2019).

To conclude, the motivation behind this project is twofold; firstly, to further advance competitive match prediction methodologies using machine learning techniques and secondly to discover valuable insights into how different chess features can influence chess games. Not only are spectators trying to predict the outcomes of games, but so are players during their preparation. It makes sense that players will base their preparation around their knowledge of how to achieve a win and to do so you must have sufficient knowledge about factors that lead to a win.

2 Aims and Objectives

2.1 Aim

- Develop a machine learning model for the prediction of chess match outcomes from chess players historical data.

2.2 Objectives

1. Choose a programming language based on available libraries for machine learning and dataset creation.
2. Determine what data is feasible for collection from online chess platforms.
3. Collect statistical and game data from 100 different players to facilitate pair-wise matchups between players in the dataset from an online chess platform.
4. Prepare dataset to ensure a suitable feature set used in the machine learning process.
5. Train two different multiclass classification machine learning models.
6. Evaluate each of the classification models accuracy and performance in predicting chess match outcomes.

3 Project plan and Risk Analysis

3.1 Project Plan

Provided within this section is a detailed project plan table highlighting the breakdown of each of the objectives set for the project's aim. Each objective has been broken down into a set of aims and in turn, a deliverable to help identify when each aim has been successfully satisfied (see **Table 1**). Furthermore, a Gant chart has been provided to help better understand the flow of the project's plan from start to finish (see **Fig. 1**).

Table 1. This project plan table contains six columns which are Objectives, Aim, Deliverables, Start Date, End Date, and No of Weeks. Each row within the table is linked back to the objectives given within section 2.2 of this proposal, for example objective 6.2 is linked directly to completing objective 6.

Objective	Aim	Deliverables	Start Date	End Date	No of Weeks
1.1	Determine programming language that will be used and available libraries.	Decision on programming language and selected libraries that will be used to create both a dataset and a machine learning model.	06/11/2023	12/11/2023	1
2.1	Find Application Programming Interfaces (APIs) for data collection.	Identified different API's that allow for the extraction of data from online chess platforms.	13/11/2023	26/11/2023	2
2.2	Investigate the available API's end points and	Understanding of what data should and can be	27/11/2023	10/12/2023	2

	capabilities to gain an understanding of what data can be collected.	collected through the previously found APIs.			
3.1	Begin using the APIs to collect any data that is deemed relevant for the dataset that will be used for training the machine learning model.	Semi-structured data acquired using previously found APIs.	11/12/2023	31/12/2023	3
3.2	Turn semi-structured historical chess data into structured data.	Historical chess data in structured format.	01/01/2024	14/01/2024	2
4.1	Carry out data cleaning to remove any inaccurate or unwanted data within the dataset and merge collected data.	Obtained cleaned dataset with no errors ready for use in the creation of the feature set.	15/01/2024	28/01/2024	2
4.2	Carry out data preprocessing and identify features that can be used or created from the structured historical chess dataset.	Created a feature set ready for training a machine learning classification model with the inclusion of labels.	29/01/2024	19/02/2024	3
5.1	Determine two different supervised machine learning algorithms that can be used in reference to the feature set and output requirements.	Two Machine learning algorithms selected for use.	19/02/2024	25/02/2024	1
5.2	Train one of the machine learning algorithms on the feature set.	One machine learning model trained on the created feature set.	26/02/2024	10/03/2024	2
5.3	Train second machine	Both machine learning models	11/03/2024	24/03/2024	2

	learning algorithm on the feature set.	trained on the created feature set.			
6.1	Evaluate Both of the machine learning models using relevant metrics.	Obtained valuable insights that can be used to evaluate both machine learning models performance.	25/03/2024	07/04/2024	2
6.2	Use acquired insights about both machine learning models to compare the difference in performance.	Identified which model performed better on the chess feature set.	08/04/2024	14/04/2024	1
6.3	Compare the results of machine learning models against other relevant machine learning models in the realm of sport and chess match outcome prediction.	Identified how well the created model performs with reference to other models within the same research area.	15/04/2024	28/04/2024	2

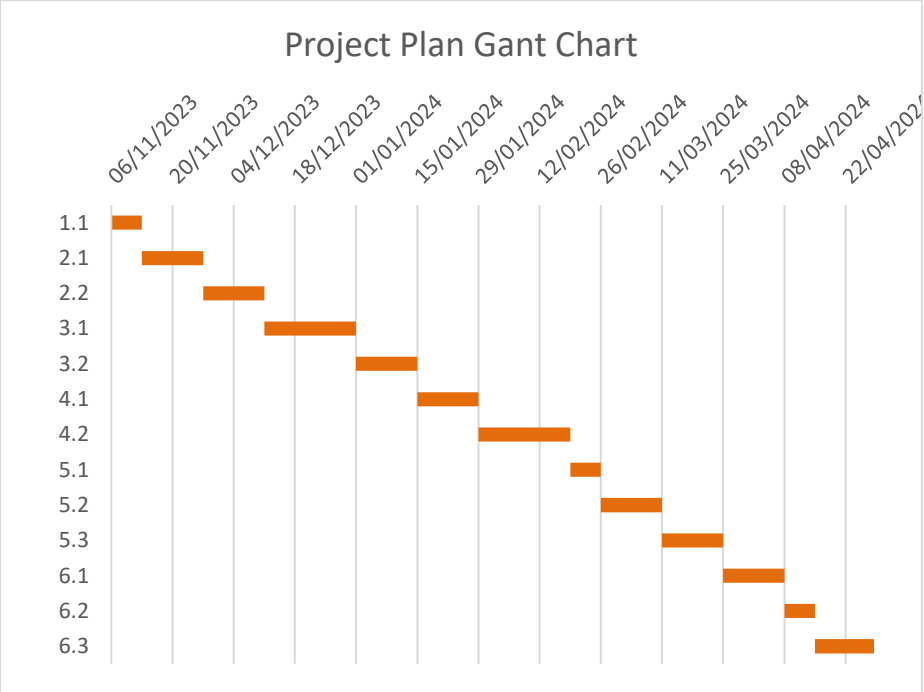

Fig. 1. The above figure is a Gant chart created in excel spreadsheet from the project plan table (see **Table 1**). The time scale of each line within the Gant chart is 14 days. For example, objective 1.1 is shown to end in between two date lines, illustrating that the time taken is 1 week or 7 days.

4 Risk Analysis

In any project it is always important to ensure that any risks that may occur during its course are anticipated, mitigated, and accounted for in the case that they do arise to ensure a smooth workflow. Included within this section is a risk matrix (see **Table 2**) followed by a risk assessment table (see **Table 3**) which contains identified risks, a risk rating that can be deduced using the probability and impact from the risk matrix, mitigation strategies and finally contingency plans to be used in the circumstances that the risk arises.

Table 2. The below table is a 5 x 5 Risk Matrix. Each row within the matrix represents a different probability from Rare to Almost certain. Each column of the matrix indicates the impact from Negligible to Severe. This matrix is used to give each risk a measured Risk Rating (see **Table 3**).

Impact		**Negligible**	**Minor**	**Significant**	**Major**	**Severe**
Probability	**Almost Certain**	Moderate	Severe	Severe	Critical	Critical
	Likely	Sustainable	Moderate	Severe	Critical	Critical
	Possible	Sustainable	Moderate	Moderate	Severe	Critical
	Unlikely	Sustainable	Sustainable	Moderate	Severe	Critical
	Rare	Sustainable	Sustainable	Sustainable	Moderate	Severe

Table 3. This Risk Assessment table has been created for the project plan (see **Table 1**). Each row of the table identifies a risk related to specific aims within the project plan. Each risk is given a Risk Rating, Mitigation Strategy and Contingency Plan.

Related Tasks	Risk	Likelihood	Impact	Risk Rating	Mitigation Strategy	Contingency Plan
1.1	Limited capability with selected programming language library.	Unlikely	Minor	Sustainable	Ensure that thorough research is carried out on the different libraries available and their capabilities before selection.	Use another library found during the library Research phase.
2.1	Unable to Find any APIs for the collection of data from online chess platforms.	Unlikely	Significant	Moderate	Ensure that each platform is investigated thoroughly for a viable API.	Acquire a dataset from the data science competition platform Kaggle.
2.1, 2.2	Unable to obtain relevant data using the APIs for the online chess platforms.	Unlikely	Significant	Moderate	Ensure that each APIs documentation is thoroughly investigated to ensure that relevant data can be collected.	Acquire a dataset from the data science competition platform Kaggle.

3.2, 4.1, 4.2	Loss of data during the transformations from unstructured data, data cleaning or data preprocessing.	Likely	Major	Critical	Implement backups of data both before and after data collection and transformation.	Return to data collection phase and re-acquire the data using previously used API endpoints.
4.2	Not enough features for the model to learn from sufficiently	Possible	Major	Severe	Carry out thorough feature engineering during data preprocessing stages to create more relevant features.	Return to data collection phase and acquire more data if possible. If this is not feasible, Acquire a dataset from Kaggle.
5.1	Limited understanding of the selected machine learning algorithm(s).	Likely	Minor	Moderate	Carry out research into each of the machine learning algorithms and relevant documentation before selection.	Choose a different algorithm which is more familiar or find some relevant tutorials to gain further knowledge.
5.2, 5.3	Time-consuming training process due to potential code adjustments to fix prior mistakes within code.	Likely	Significant	Severe	Seek information from online documentation to bolster knowledge. Furthermore ensure a methodical approach and break down tasks during code implementation.	Allocate extra time in the project to account for difficulties during code implementation and seek supervisors' guidance.
6.1, 6.2, 6.3	Limited understanding of the required metrics to measure machine learning algorithms performance.	Possible	Minor	Moderate	Research various metrics that are applicable to different machine learning algorithms from online sources and academic literature.	Engage with supervisor or seek guidance from lecturers experienced with machine learning.

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10

Word Count

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